

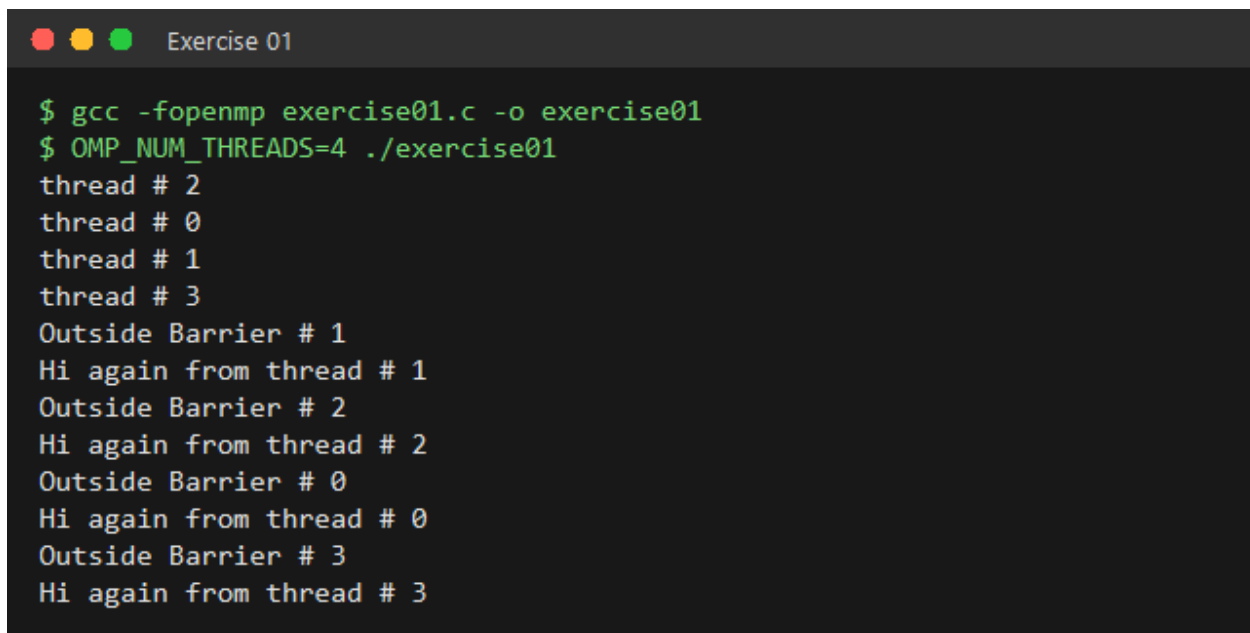
EXERCISE 1 - #pragma omp barrier

Files : exercise01.c

Compile / run

```
gcc -fopenmp exercise01.c -o exercise01
```

```
OMP_NUM_THREADS=4 ./exercise01
```



```
Exercise 01

$ gcc -fopenmp exercise01.c -o exercise01
$ OMP_NUM_THREADS=4 ./exercise01
thread # 2
thread # 0
thread # 1
thread # 3
Outside Barrier # 1
Hi again from thread # 1
Outside Barrier # 2
Hi again from thread # 2
Outside Barrier # 0
Hi again from thread # 0
Outside Barrier # 3
Hi again from thread # 3
```

How omp barrier works in this program

- * #pragma omp parallel creates a team of 4 threads; each prints "thread # n".
- * #pragma omp barrier is a synchronisation point. No thread may pass the barrier until EVERY thread in the team has reached it.
- * Because of this, all four "thread # n" lines are printed BEFORE any "Outside Barrier" / "Hi again" line - the two groups never mix.
- * Inside each group the order is still random, since the threads run independently once they are released from the barrier.